Worksheet No. - 05

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Semester: 3rd **Date of Performance: 09.10.2026**
Subject Name: Automation Software Testing **Subject Code: 25CAH-705**

Aim/Overview of the practical: Design and implement automated test scripts using Playwright to validate the login functionality and form input fields of a web application by verifying user authentication, input validations, and appropriate system responses for both valid and invalid scenarios.

Objective: To design and implement automated test scripts using Playwright to verify login functionality, validate form input fields, and check the system's response to both valid and invalid credentials.

Code

```
import asyncio
import os
from playwright.async_api import async_playwright, expect

url = "https://www.saucedemo.com/"

os.makedirs("screenshots", exist_ok=True)

async def take_screenshot(page, name):
    await page.screenshot(
        path=f"screenshots/{name}.png",
        full_page=True
    )

async def test_login():
    async with async_playwright() as p:
        browser = await p.chromium.launch(
            headless=False,
            slow_mo=500
        )

        page = await browser.new_page()
        page.set_default_timeout(15000)

        # TEST 1: Valid login
        print("\nTEST 1: Valid Login")

        await page.goto(url)
```

```
await page.locator("#user-name").fill("standard_user")
await page.locator("#password").fill("secret_sauce")
await take_screenshot(page, "01_valid_credentials")

await page.locator("#login-button").click()
await expect(page).to_have_url(
    "https://www.saucedemo.com/inventory.html"
)
await expect(page.locator(".title")).to_have_text("Products")

await take_screenshot(page, "02_login_success")
print("PASS: Valid login successful.")

await page.wait_for_timeout(1500)

# TEST 2: Invalid username
print("\nTEST 2: Invalid Username")

await page.goto(url)
await page.locator("#user-name").fill("wrong_user")
await page.locator("#password").fill("secret_sauce")
await take_screenshot(page, "03_invalid_username_input")

await page.locator("#login-button").click()
await expect(page.locator("[data-test='error']")).to_be_visible()
await take_screenshot(page, "04_invalid_username_error")

print("PASS: Invalid username rejected.")
await page.wait_for_timeout(1500)

# TEST 3: Invalid password
print("\nTEST 3: Invalid Password")

await page.goto(url)
await page.locator("#user-name").fill("standard_user")
await page.locator("#password").fill("wrong_password")
await take_screenshot(page, "05_invalid_password_input")

await page.locator("#login-button").click()
await expect(page.locator("[data-test='error']")).to_contain_text(
    "Username and password do not match"
)
await take_screenshot(page, "06_invalid_password_error")

print("PASS: Invalid password rejected.")
await page.wait_for_timeout(1500)

# TEST 4: Empty fields
print("\nTEST 4: Empty Fields")

await page.goto(url)
```

```
    await page.locator("#user-name").fill("")
    await page.locator("#password").fill("")
    await take_screenshot(page, "07_empty_fields")

    await page.locator("#login-button").click()
    await expect(page.locator("[data-test='error']")).to_contain_text(
        "Username is required"
    )
    await take_screenshot(page, "08_empty_fields_error")

    print("PASS: Empty fields rejected.")
    await page.wait_for_timeout(1500)

    # TEST 5: Empty password
    print("\nTEST 5: Empty Password")

    await page.goto(url)
    await page.locator("#user-name").fill("standard_user")
    await page.locator("#password").fill("")
    await take_screenshot(page, "09_empty_password")

    await page.locator("#login-button").click()
    await expect(page.locator("[data-test='error']")).to_contain_text(
        "Password is required"
    )
    await take_screenshot(page, "10_empty_password_error")

    print("PASS: Empty password rejected.")

    print("\nAll 5 login tests passed!")
    print("Check the screenshots folder for all screenshots.")

    await page.wait_for_timeout(3000)
    await browser.close()

asyncio.run(test_login())
```

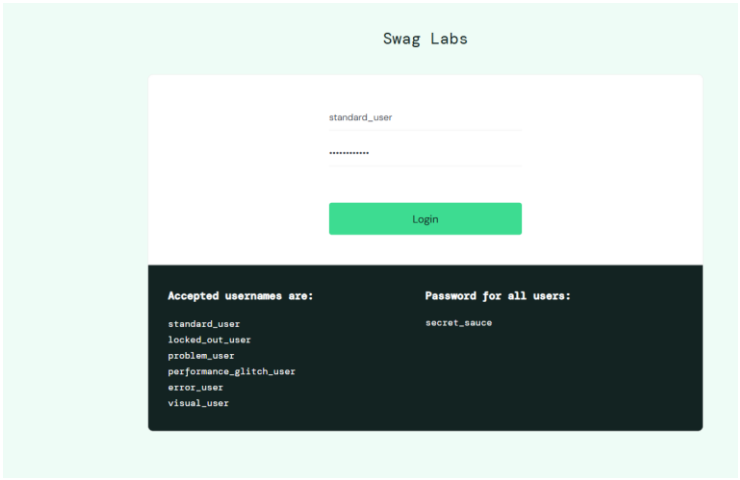
Test Cases

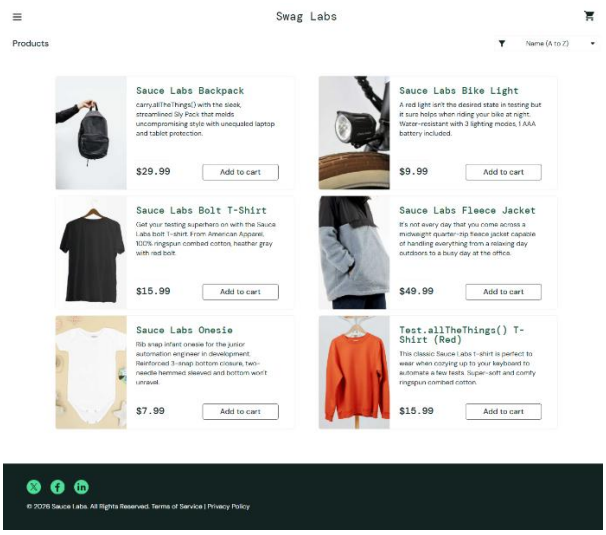
Test Case ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC01	Login with valid credentials	Username: standard_user Password: secret_sauce	User logs in successfully and the Products page is displayed.	User is redirected to the Products page, and the heading “Products” is displayed.	Pass

Test Case ID	Test Scenario	Test Data	Expected Result	Actual Result	Status
TC02	Login with invalid username	Username: wrong_user Password: secret_sauce	An error message is displayed, and login is denied.	Error message indicates that the username and password do not match.	Pass
TC03	Login with invalid password	Username: standard_user Password: wrong_password	An error message is displayed, and login is denied.	Error message indicates that the username and password do not match.	Pass
TC04	Login with both fields empty	Username: blank Password: blank	A username-required error message is displayed.	Error message displays “Username is required.”	Pass
TC05	Login with empty password	Username: standard_user Password: blank	A password-required error message is displayed.	Error message displays “Password is required.”	Pass

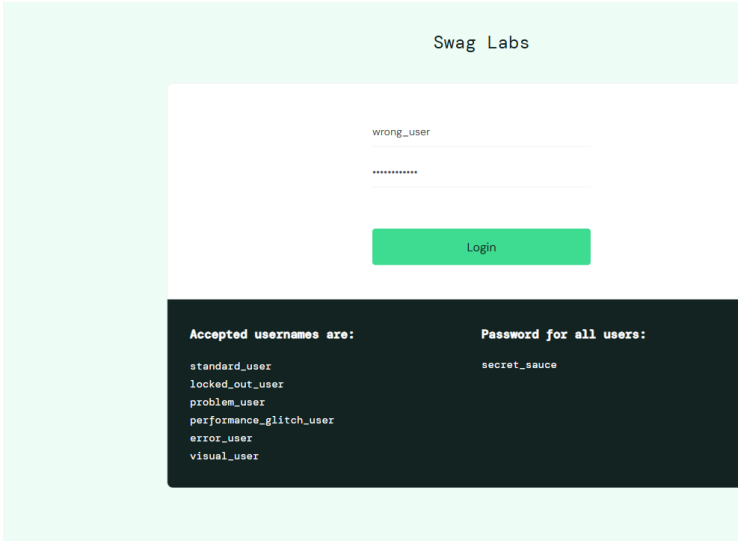
Output

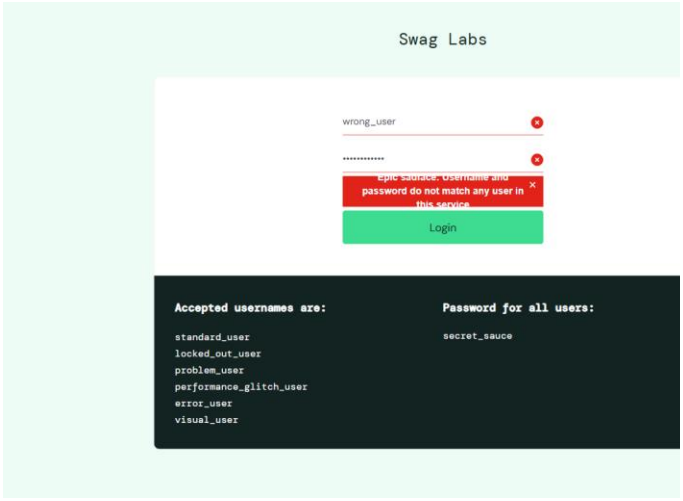
Valid Test Cases



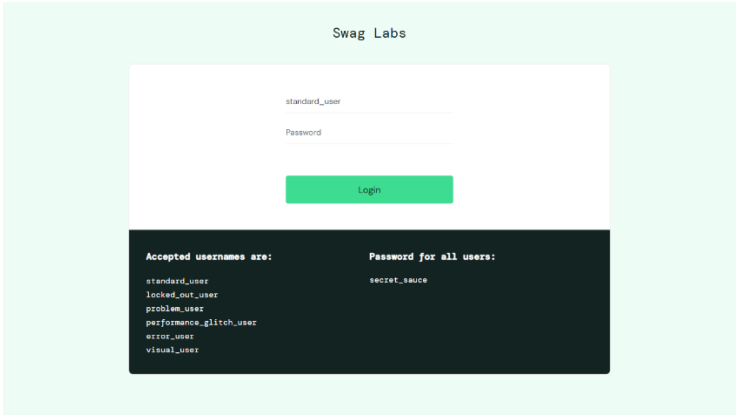


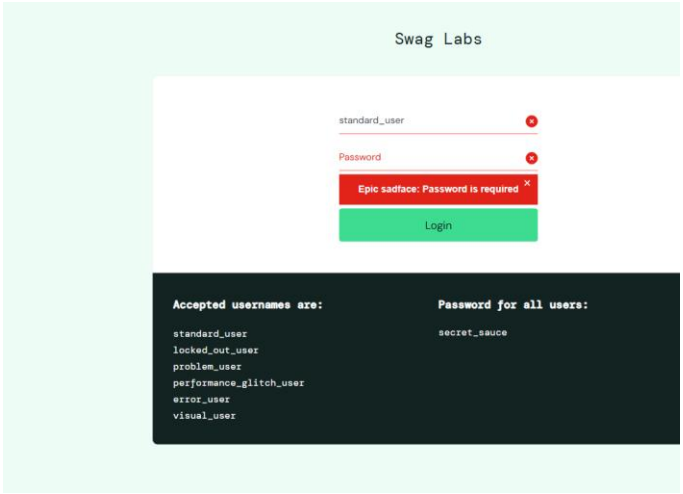
Invalid Test Cases





Empty Test Cases





Learning outcomes (What I have learnt):

1. Learned to create and execute automated test scripts using Playwright.
2. Understood how to test login functionality with valid and invalid credentials.
3. Learned to validate form input fields and error messages.
4. Gained knowledge of verifying expected and actual system responses.

Evaluation Grid:

Sr. No.	Parameters	Marks Obtained	Maximum Marks
1.	Demonstration and Performance		12
2.	Worksheet		8
3.	Viva		10